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Lifestyle Advice Model:
Using a semantic model to support personalised lifestyle advice

Introduction

This project examines whether semantic models can enhance the accuracy of personalized lifestyle advice by linking biomedical and lifestyle data to evidence-based knowledge.

A semantic model is used to represent individuals, their behaviours, and clinical measurements, and to connect these data to lifestyle advice through defined relationships and trigger conditions derived from peer-reviewed studies.

This approach enables the structured evaluation of personal data against curated knowledge, supporting a more consistent and interpretable generation of lifestyle advice.

Workflow

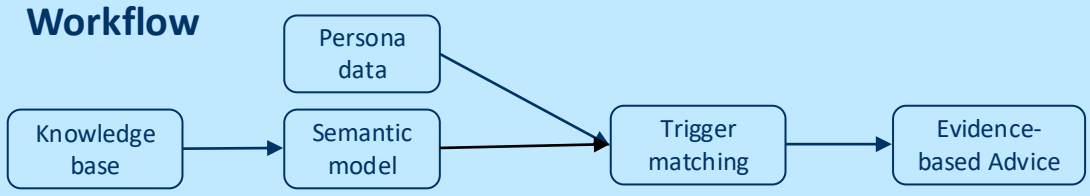

Figure 1. Workflow for generating personalized lifestyle advice using a semantic model and a curated knowledge base.

From data to advice

- The system integrates a **semantic model** with a curated **knowledge base**
- **Persona data** is mapped to the model
- Data is evaluated against **trigger conditions** derived from peer-reviewed studies
- Matching conditions trigger corresponding **evidence-based lifestyle advice**

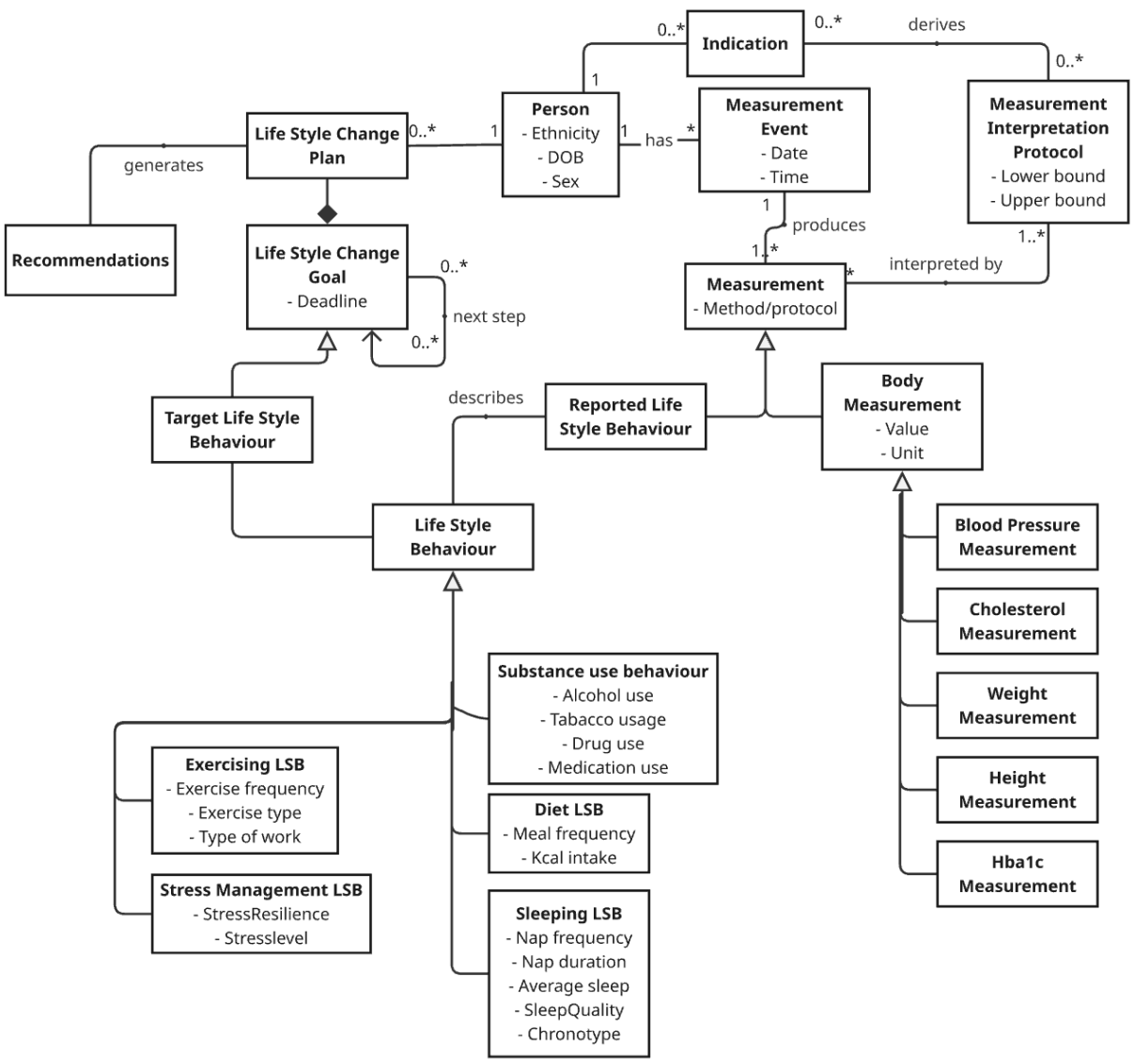

Figure 2. Semantic model representing relationships between lifestyle behaviours, biomedical measurements and lifestyle recommendations.

Semantic model

The semantic model represents the relationships between a person, their lifestyle behaviours, and biomedical measurements within a structured framework.

The model was implemented in Protégé using ontology-based representations to structure biomedical and lifestyle data.

Persona data is instantiated in the model and linked to behavioural indicators and clinical measurements, such as physical activity, diet, and biomarkers.

These data points are evaluated against trigger conditions derived from peer-reviewed studies, which activate corresponding evidence-based lifestyle recommendations.

Results

Ontology Mapping: Integrated 8 core ontologies, leveraging SNOMED CT for clinical data and BCIO for behavior change logic.

High Performance: Advice generation executed in ~0.23 seconds per persona.

Validated Logic: Persona data successfully triggered evidence-based advice when conditions from knowledge bases were met.

References

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Take Home Message

A semantic model provides a structured framework for integrating biomedical and lifestyle data with evidence-based knowledge, supporting automated lifestyle advice generation.